3053994 || BIOL5177 || Assignment 2

Task 1

Figure 1. IGV view of *PF3D7_0102800* from the *Plasmodium falciparum* 3D7 (v24) genome based on 10x Genomics 3' scRNA-seq data. Tracks show read coverage, splice junctions, and aligned reads. The snapshot shows high read coverage, indicating strong expression. The junction track reveals multiple splicing events, indicating transcript diversity. Interestingly, although this is a 3' chemistry 10x run, the coverage peak is at the 5' end. One possible explanation is internal priming or incomplete reverse transcription, causing reads to accumulate upstream of the true 3' end.

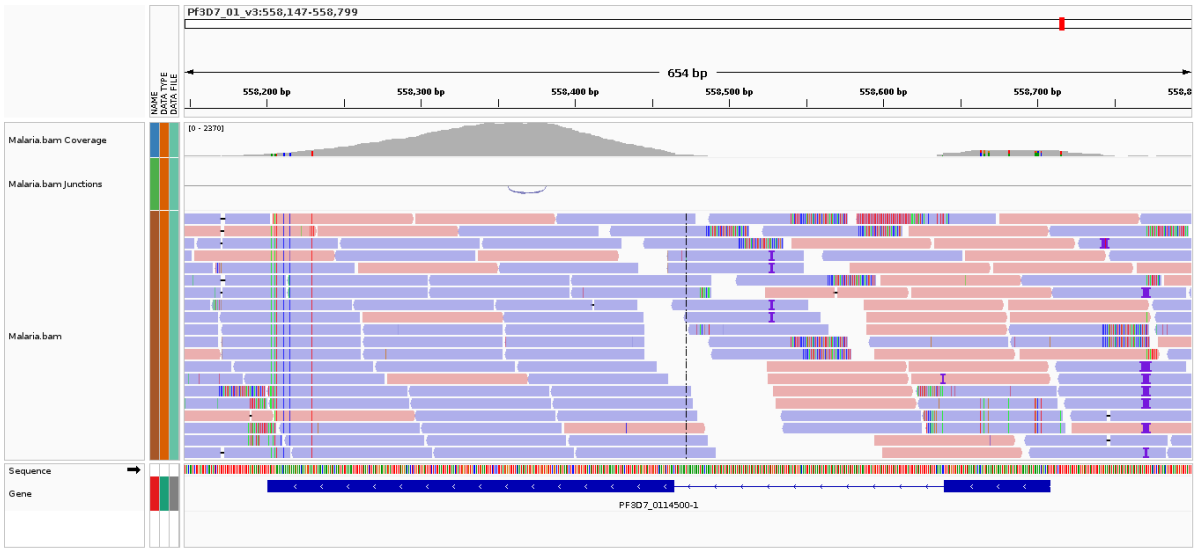


Figure 2. IGV view of *PF3D7_0114500* from the *Plasmodium falciparum* 3D7 (v24) genome based on 10x Genomics 3' scRNA-seq data. The tracks display read coverage, splice junctions, and aligned reads. IGV snapshot of gene PF3D7_0114500 shows high read coverage and clear support for alternative splicing. Two exons are observed with strong

peaks at both ends, and splice junction reads confirm the presence of exon-intron boundaries. These findings align well with the gene model and suggest that PF3D7_0114500 undergoes regulated splicing, possibly producing multiple isoforms in *Plasmodium falciparum*.

Task 2 (a)

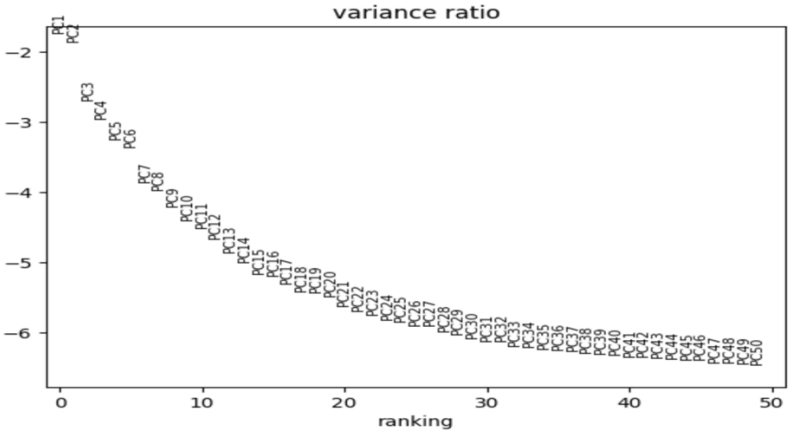
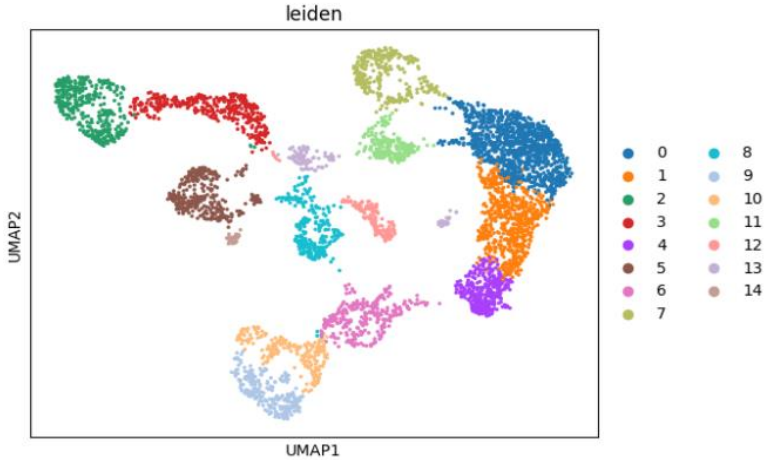


Figure 1. Principal component analysis (PCA) was performed for dimensionality reduction. I selected up to 50 PCs as the variance ratio plot above shows a gradual elbow extending up to PC50, suggesting continued contribution to meaningful variation.

Task 2 (b)



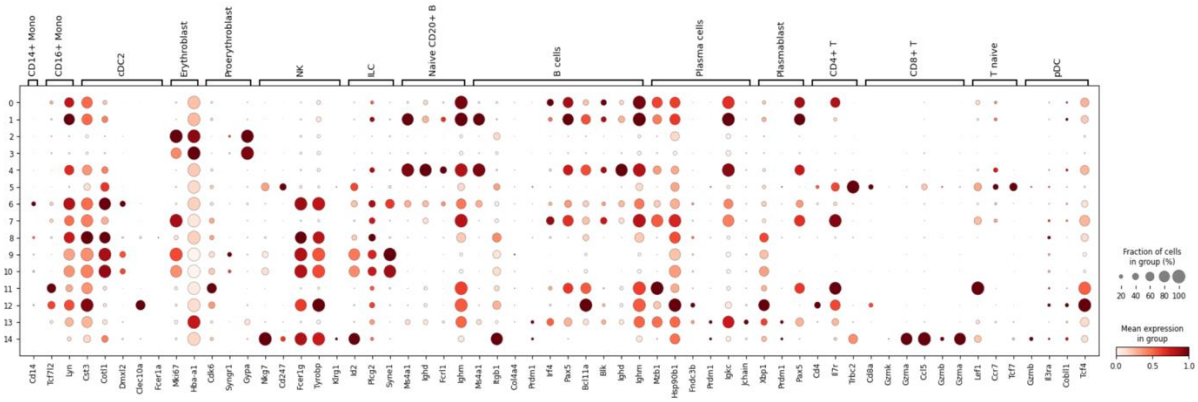


Figure 2. UMAP of BKNN-integrated data clustered using Leiden resolution 0.50 (top panel), which yielded 14 clusters — a balance between overly coarse and overly fragmented. Clusters are ordered by size (0 = largest, 14 = smallest). Based on marker gene expression, the top 3 largest clusters were annotated as:

- Cluster 0: B cells
- Cluster 1: Naive CD20+ B cells
- Cluster 2: Erythroblasts

These annotations were added to the concatenated adata object (see Scanpy script). The clustering captures major cell types with biologically meaningful resolution.

The dot plot (bottom panel) shows the expression of key marker genes across all clusters. Darker red dots mean a gene is more highly expressed in that cluster compared to the others, and larger dots indicate more cells in that cluster express the gene. From this, we can clearly see that:

- Cluster 0 has strong relative expression of B cell markers
- Cluster 1 shows higher relative expression of naive CD20+ B cell markers
- Cluster 2 shows elevated expression of erythroblast markers

These patterns help confirm the cell type identities of the top three clusters based on both marker gene expression and the proportion of cells expressing them.

<u>Task 2 (c)</u>

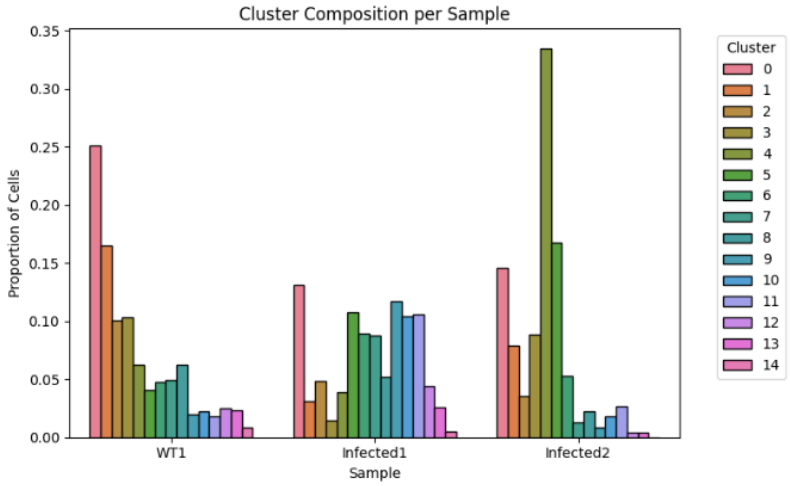


Figure 3. The bar plot shows the proportion of cells from each cluster across samples. Clusters 0 and 1, both B cell populations (with cluster 1 being Naive CD20+ B), are reduced in infected samples, suggesting a loss of these B cell types during malaria infection. Interestingly, cluster 4, which shares a similar B cell expression profile with cluster 1 (including Ms4a1), is elevated in Infected2. However, Infected2 has a low total cell count (227 cells post-filtering), which may exaggerate proportional changes. This highlights a key limitation: frequency comparisons can be skewed by unequal sample sizes or batch effects.

Additionally, BKNN integration modifies only the neighborhood graph without adjusting the gene expression matrix, meaning it can correct batch effects in clustering or UMAP, but may not fully account for technical variation in downstream analyses like differential expression. In some cases, BKNN may also overcorrect or obscure true biological differences if they are confounded with batch structure.

<u>Task 2 (d) (e) (f)</u>

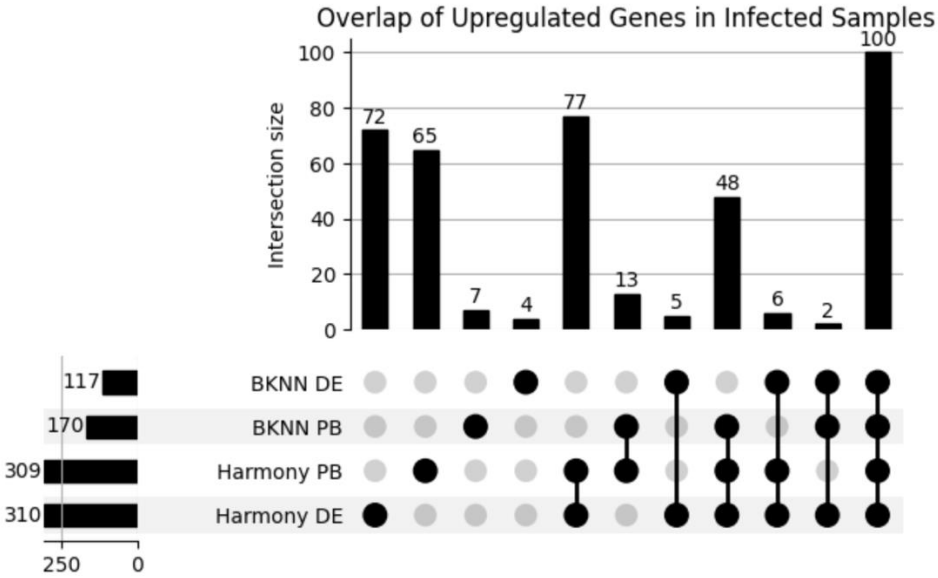


Figure 4. UpSet plot showing the overlap of genes upregulated in infected samples across four differential expression (DE) methods:

- BKNN DE
- BKNN pseudo-bulk (PB)
- Harmony PB
- Harmony DE

The horizontal bar plots on the lower left represent the total number of upregulated genes identified by each method individually:

- 117 genes from BKNN DE
- 170 genes from BKNN PB
- 309 genes from Harmony PB
- 310 genes from Harmony DE

The vertical bar plots at the top indicate the size of each intersection between methods. Each vertical bar corresponds to a specific combination of methods, shown by the black dots underneath:

- A single black dot (e.g., on "Harmony DE" only) means genes found only in that method — for example, 72 genes are unique to Harmony DE.

- Connected black dots (e.g., all four rows) show shared genes — for instance, 100 genes are common across all four methods.
- Other bars show partial overlaps, like 65 genes found only in Harmony PB, or 7 found only in BKNN PB.

This plot helps visualize both method-specific genes and the core set of consistently detected genes shared across multiple approaches.

Task 2 (f)

Harmony integrates data by adjusting the principal component (PC) embeddings, directly correcting batch effects in the reduced dimensional space. This often leads to strong alignment across batches and allows for better integration in downstream clustering. Harmony tends to be more sensitive, identifying a larger number of differentially expressed genes. However, this sensitivity may come at the cost of increased false positives or overcorrection, potentially masking true biological differences if batch and condition are confounded.

BKNN, on the other hand, modifies the neighborhood graph to connect similar cells across batches without altering the underlying expression matrix. This helps preserve biological variation and works well with UMAP and clustering.

To improve statistical power in differential expression analysis — especially important as only one WT and two infected samples were available — I used a pseudo-replicate approach in the BKNN (PB) and Harmony (PB) pipelines. This aggregation strategy reduces single-cell noise and allows the use of robust bulk-like statistical models, which may explain why BKNN pseudo-bulk detected more DE genes than the single-cell DE analysis.

Conclusion:
Each method contributes complementary strengths. To balance sensitivity and reliability, I would select the 100 genes consistently upregulated across all four methods as a high-confidence set for downstream interpretation.

Task 2 (g)

The cells in the dataset most likely come from the spleen, a central immune organ where diverse immune cell types — including T cells, B cells, NK cells, monocytes, and plasmablasts — coexist alongside erythroid progenitors. The dot plot in Figure 2 confirms the presence of these populations, consistent with splenic architecture. In the context of malaria infection, the spleen plays a key role in both immune activation and stress erythropoiesis, making it a biologically plausible source of the sample. The combination of developing erythroid cells and activated effector cells supports this conclusion.

Task 3

When Ms4a1 cluster is assumed as **end** point:

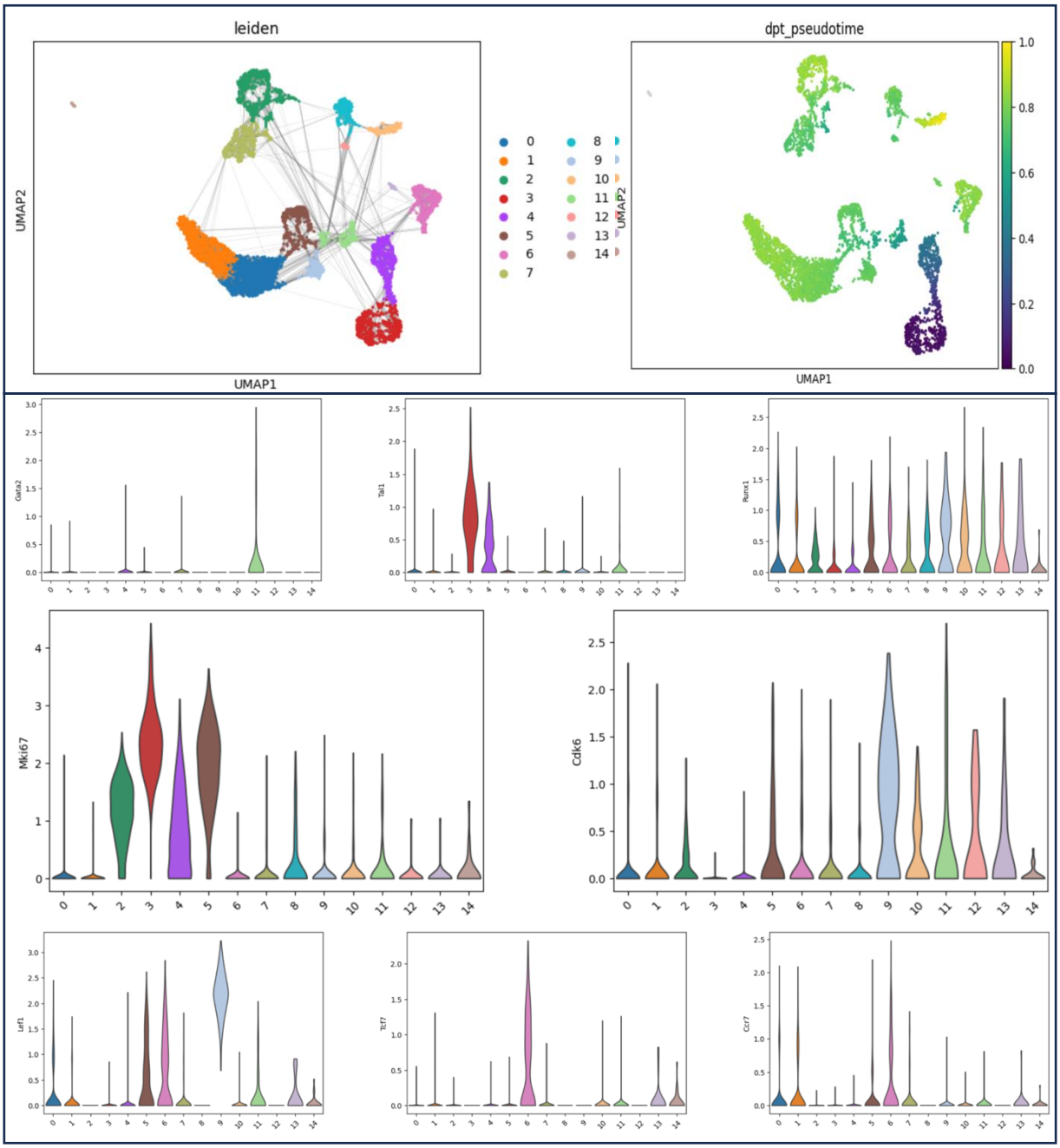


Figure 5.
Pseudotime analysis was performed to infer cellular progression in the malaria-infected spleen-like tissue. The UMAP (top left) shows distinct clusters with inferred connectivity lines, where cluster 1 (orange) represents the Ms4a1-expressing B cell cluster, our main cluster of interest. To identify the starting point of the trajectory, gene expression patterns associated with early progenitor and proliferative states were examined.Violin plots (bottom panels) show that cluster 3 (red) has elevated expression of hematopoietic progenitor markers(*Gata2*, *Tal1*, *Runx1*) and proliferation-associated genes (*Mki67*, *Cdk6*), suggesting it represents the root of pseudotime. Additional expression of early T cell markers (*Lef1*, *Tcf7*, *Ccr7*) in the lower panel supports this as a multipotent or early lymphoid state.The pseudotime UMAP (top right) shows the inferred developmental progression of cells. Based on the expression of these early markers, we manually selected cluster 3 as the pseudotime root. The pseudotime values reflect this assignment, with cluster 3 at the early end of the trajectory. Cluster 1 (Ms4a1 B cells) appears at an intermediate stage, suggesting a transitional or activated B cell population. The trajectory terminates at cluster 10, which expresses a mix of effector markers and likely contains terminally differentiated immune cells (e.g., NK cells, DCs, and mature B cells), as supported by the earlier dot plot in Figure 2.

When Ms4a1 cluster is assumed as **start** point:

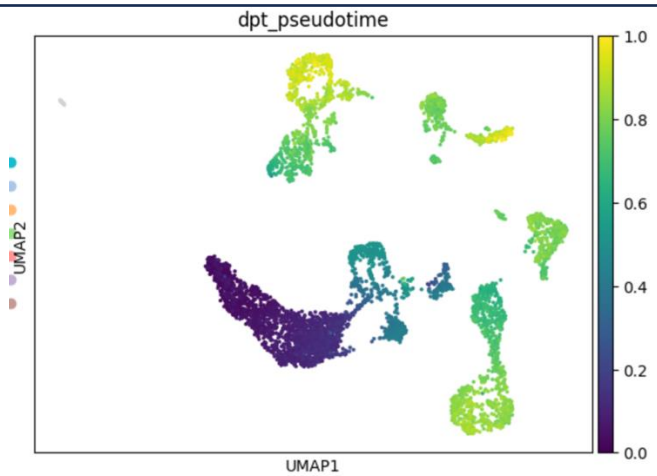


Figure 6.
Pseudotime analysis using UMAP embedding with cluster 1 (Ms4a1+ B cells) manually set as the starting point. The trajectory begins in this B cell population (low pseudotime values, dark purple) and progresses toward cluster 10, which was identified by PAGA as the likely terminal state. Based on marker gene expression in earlier analyses (Figure 2), cluster 10 expresses a combination of effector genes associated with NK cells, dendritic cells (DCs), and mature B cells, suggesting it represents a terminally differentiated or activated immune compartment.

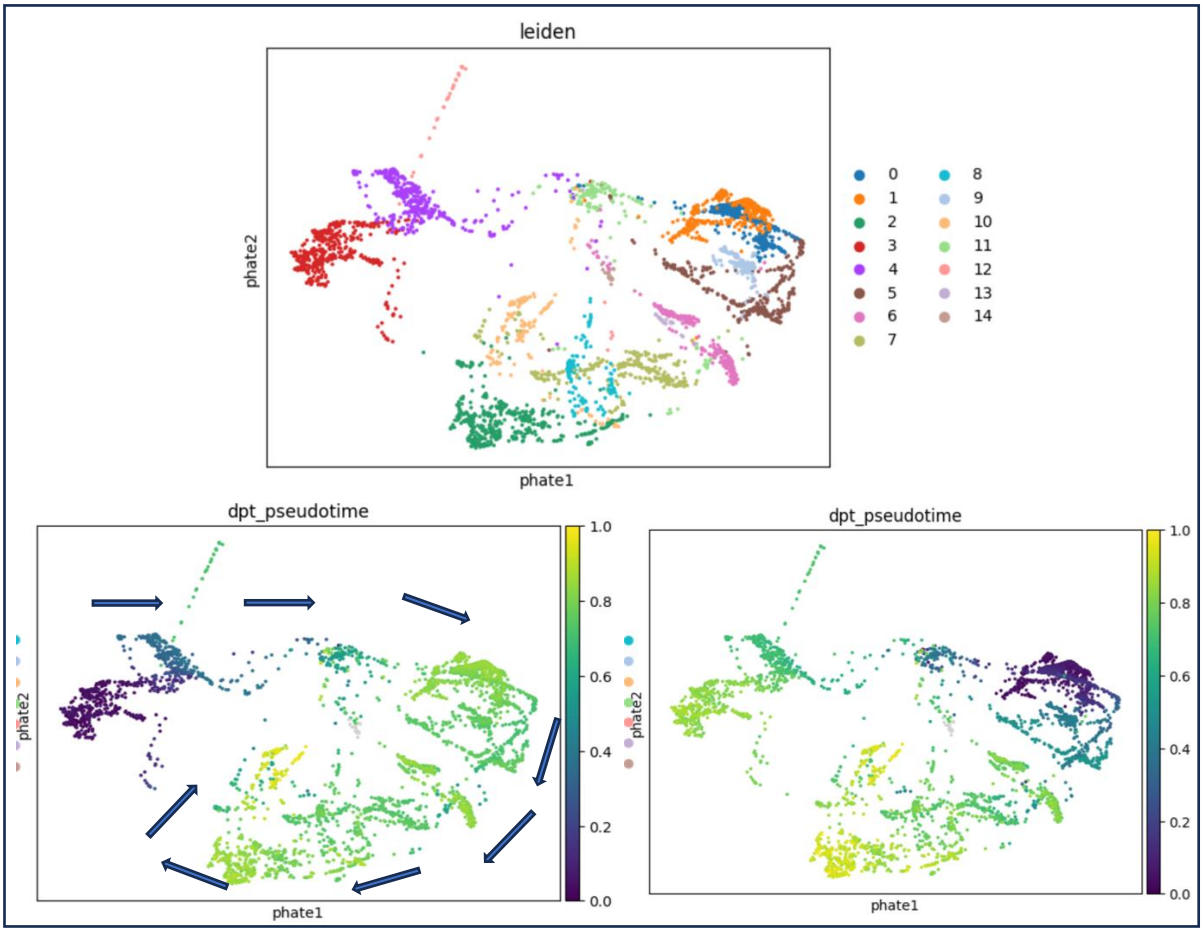


Figure 7.
PHATE embedding was used to visualize cellular trajectories, providing a smoother and more continuous representation of lineage relationships compared to UMAP. The top panel shows cluster identities on the PHATE map, with cluster 3 (red) and cluster 1 (orange, Ms4a1+ B cells) occupying distinct regions.

In the lower left panel, cluster 3 was set as the pseudotime root, revealing a trajectory from early progenitors toward more differentiated immune states. In the lower right panel, the trajectory starts from cluster 1, highlighting a B cell activation and differentiation pathway. Notably, PHATE reveals a circular or looping trajectory (demonstrated with arrows)—suggesting that cells may progress through states and potentially return to a central immune-regulatory or transitional state.

Overall, PHATE appears to better capture the global structure and complexity of the immune landscape in this dataset, particularly under malaria-induced conditions, where dynamic cell transitions and activation cycles are expected.

Discussion for task 3 (~200 words)

Figures 5–7. Pseudotime analysis was performed to infer cellular progression in a malaria-infected spleen-like tissue. Two scenarios were explored using UMAP and PHATE embeddings: one where cluster 3 (red, enriched for *Gata2*, *Tal1*, *Runx1*, *Mki67*) is set as the starting point, and one where cluster 1 (orange, *Ms4a1*+ B cells) is the start.

In the first scenario (**Figure 5**), gene expression of hematopoietic progenitor and proliferation markers in cluster 3 supports it as an early developmental state. Additional markers (*Lef1*, *Tcf7*, *Ccr7*) suggest a multipotent or early lymphoid identity. The pseudotime trajectory shows progression from cluster 3 to cluster 10, which expresses a mix of effector genes (e.g., from NK cells, DCs, and mature B cells), suggesting it is a likely terminal state. PAGA did not assign the Ms4a1 cluster as terminal, reinforcing its likely role as a transitional or activated B cell state rather than a final differentiation point.

In the second scenario (**Figure 6**), cluster 1 (*Ms4a1*+ B cells) was manually set as the root, modeling a B cell activation and differentiation pathway. The trajectory again ends at cluster 10, highlighting its consistent role as a terminal immune state across both lineage-based and activation-based trajectories. This contrast illustrates two distinct biological interpretations: one representing a classical hematopoietic progression (cluster 3 as start), and the other reflecting a B cell–driven immune response (cluster 1 as start).

Figure 7 presents the same analysis using PHATE embedding, which provides a smoother and more continuous view of cellular transitions. PHATE further highlights a circular or looping trajectory (see arrows in PHATE embedding graph). Overall, PHATE better captures the global structure and complexity of the immune landscape under infection conditions.